

# Chapter 8

## AI in Sexual and Domestic Partner Violence



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### 1 Introduction

Sexual and intimate partner violence (S/IPV) are persistent and complex global public health problems, and they are violations of human rights. The term “Sexual Violence” (SV) comprises numerous acts that are committed or attempted by another person without freely given consent of the victim or against someone unable to consent or refuse, including penetration, forced touching, non-contact sexual acts, and situations where the victim is compelled to engage with a third party, whether through force, intoxication, or nonphysical pressure (Basile et al., 2014). It can be committed by any perpetrator (e.g., stranger, friend, person in a position of power, family member, or intimate partner) and occurs in and outside of the context of intimate partner violence (Basile et al., 2022). Intimate Partner Violence (IPV) is a form of violence that occurs between a current or former intimate partner (i.e., spouse, boyfriend/girlfriend, dating partner, or ongoing sexual partner) and includes physical violence, sexual violence, stalking, and psychological aggression (including coercive tactics) (Breiding et al., 2015). IPV may also vary in severity or duration and does not require sexual intimacy (Leemis et al., 2022). Both forms of violence transcend cultural, economic, and geographic boundaries, affecting individuals across the lifespan and perpetuating cycles of trauma within communities. This chapter will primarily discuss these two forms of violence collectively, with special notes when they are discussed separately.

The personal and societal impact of S/IPV is profound and multifaceted. Beyond the short-term physical injuries and death, S/IPV contributes to long-term health

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consequences such as difficulty with daily activities, memory loss, chronic pain, gastrointestinal disorders, gynecological signs like transmitted diseases, and mental health conditions, including depression, anxiety, post-traumatic stress disorder (PTSD), and suicidal thoughts (Campbell, 2002; Ellsberg et al., 2008). Also, IPV has a negative physical and emotional impact on family members, especially children (Chan et al., 2021). Econometric methods, including direct medical costs and indirect productivity losses, have quantified these health consequences. Peterson et al. (2018) estimated the lifetime cost of IPV at \$103,767 per female victim and \$23,414 per male victim, with a total economic burden for the population of nearly \$3.6 trillion over victims' lifetimes. Medical costs (59%, \$2.1 trillion), lost productivity (37%, \$1.3 trillion), criminal justice activities (2%, \$73 billion), and other costs, including property loss or damage (2%, \$62 billion), comprised the total lifetime economic burden, with 37% (\$1.3 trillion) paid by government sources. Similarly, developing countries bear IPV costs equivalent to 1%–2% of GDP (Duvvury, 2013). These impacts highlight the importance of intervention and prevention of S/IPV.

Moreover, substantial data indicate that the number of individuals victimized by S/IPV is exceedingly large, with its impact extending across all countries worldwide. A World Health Organization-led analysis (WHO, 2021) of prevalence data from 2000 to 2018 across 161 countries and areas for the UN Interagency Working Group on Violence Against Women revealed that globally, approximately 30% of women have experienced physical and/or sexual violence by an intimate partner or non-partner sexual violence or both. According to the statistics discovered by The National Intimate Partner and Sexual Violence Survey (2016–2017) (Basile et al., 2022; Leemis et al., 2022) in the United States, almost 1 in 2 women and more than 2 in 5 men reported experiencing contact with sexual violence, physical violence, and/or stalking victimization by an intimate partner at some point in their lifetime; meanwhile, 1 in 4 women reported completed or attempted rape during her lifetime, and 1 in 9 men reported being made to penetrate someone during his lifetime, highlighting the pervasive nature of this public health crisis.

During the COVID-19 pandemic, IPV escalated due to measures like social distancing and shelter-in-place orders, particularly among marginalized groups. Collecting IPV data has been challenging, with traditional sources often underestimating the true scale of victimization. Amid these challenges, social media platforms like Reddit have emerged as crucial alternatives, offering IPV survivors a safe space to share experiences and seek support anonymously. Researchers (Buchner et al., 2023; Hodgkinson et al., 2023; Kim et al., 2023; Manzoor et al., 2022; Oyeboode et al., 2021) have leveraged these platforms to collect data, thereby gaining a more comprehensive understanding of the situations of IPV victims. This has not only highlighted the potential of social media in IPV surveillance and intervention but also laid the groundwork for future AI-related IPV research.

Although there is growing recognition of the seriousness of S/IPV, researchers and practitioners still face significant barriers. Generating robust evidence, such as randomized controlled trials, is hampered by funding limitations and challenges in accessing sensitive data (Taft & Campbell, 2024). At the same time, practitioners

must implement survivor-centered and trauma-informed care in settings that often lack resources, especially in marginalized communities affected by poverty, racism, and gender inequality. Prevention efforts, while well-intentioned, are frequently fragmented and not fully integrated into public health strategies (Niolon et al., 2017).

In this evolving landscape, artificial intelligence (AI) offers a transformative opportunity to address these challenges. Because of its scalability, AI can provide solutions across different populations and locations without requiring additional human resources. AI systems can rapidly process large volumes of data, identify patterns, and predict risks with impressive speed and accuracy, offering insights that strengthen current support systems and enhance the efforts of S/IPV researchers and practitioners. However, to realize this potential, careful attention must be paid to ethical design and practical implementation to avoid worsening existing structural or systemic issues. AI deployment should always be grounded in a clear understanding of its benefits and limitations, and be accompanied by ongoing efforts in policy, practice, and advocacy.

This chapter examines how AI technologies can be integrated to support survivors of S/IPV. We discuss key AI applications, including risk detection, surveillance, prediction, intervention tools, AI chatbots, and survivor empowerment, and explore their practical value for both research and practice. Alongside the potential benefits, we address major ethical issues, including privacy, bias, re-traumatization, and transparency. Throughout, we emphasize the importance of trauma-informed, survivor-centered approaches and the need for ethical frameworks to guide responsible AI adoption in this sensitive field.

## **2 AI Applications in S/IPV Response**

The integration of AI into support systems for survivors of S/IPV represents a transformative shift in addressing this pervasive societal issue. AI now plays a critical role across multiple phases of S/IPV support, from prevention and intervention to long-term survivor assistance. Drawing on evidence from sources such as social media, clinical records, telehealth, and legal documents, we explore how these technologies can enhance services while also considering important ethical and practical implications.

### ***2.1 Risk Detection, Surveillance, and Prediction***

Recent advances in natural language processing (NLP) have enabled innovative approaches to identifying risks associated with S/IPV through non-traditional data sources. This section presents relevant research findings aligned with the technological objectives, progressing from risk detection to monitoring and prediction. The discussion transitions from identifying existing incidents to forecasting future

risks, structured to illustrate how these technologies systematically address challenges in S/IPV risk assessment. This approach aims to provide readers with a clear understanding of the technological progression in resolving complex issues within this domain. Nevertheless, it's important to note that while these categories are distinct, some AI applications may span multiple stages. For example, a system that detects instances of violence on social media can also contribute to risk surveillance by tracking the frequency and nature of such mentions over time.

### **2.1.1 Risk Detection: Identifying Threats Across Data Sources**

Risk detection in the context of S/IPV refers to the use of AI to identify current or recent individual-level instances of S/IPV. This involves analyzing data to pinpoint situations where violence is occurring, has recently occurred, or where individuals are currently experiencing abuse or are in immediate danger. These tools are designed not to replace human decision-making, but to complement it by helping service providers, healthcare professionals, and researchers detect violence that might otherwise remain hidden or underreported.

The detection of S/IPV relies heavily on advanced AI methodologies, particularly NLP and machine learning. NLP techniques, such as keyword extraction, topic modeling (e.g., Latent Dirichlet Allocation, LDA), and word embeddings (e.g., Doc2Vec, Word2Vec), are instrumental in analyzing unstructured text data from diverse sources, including social media, news articles, and electronic health records (EHRs). These methods enable the identification of linguistic patterns and contextual cues that may indicate violence. Supervised machine learning models, such as logistic regression and transformer-based models (e.g., BERT, BioBERT, RoBERTa), are trained on labeled datasets to classify text or encounters as related to S/IPV. These models can achieve high accuracy in detecting violence-related content, as demonstrated in several studies. Additionally, rule-based systems, which rely on predefined lists of keywords or patterns, provide a structured approach to identifying potential cases of violence, particularly in clinical or crisis response settings.

For healthcare providers, risk-detection systems embedded in EHRs or chat-based platforms can help detect risks early. For instance, Botelle et al. (2022) evaluated an NLP model (BioBERT) for extracting violence-related information from EHRs. The model demonstrated high precision (89%–98%) and recall (89%–97%), underscoring its effectiveness in healthcare environments. Similarly, Tabaie et al. (2022) developed an NLP algorithm for identifying IPV cases in emergency department (ED) settings using predefined situational terms. The algorithm achieved 99.5% precision in detecting IPV cases, demonstrating the value of unstructured EHR data in identifying socially stigmatized conditions. These tools can serve as passive alerts for clinicians—flagging at-risk patients even when violence is not explicitly disclosed—thus enabling timely inquiry or referrals.

In crisis response settings, AI-enhanced chat platforms can help reduce delays in identifying urgent cases. Swaminathan et al. (2023) developed the Crisis Message Detector-1 (CMD-1), a two-stage NLP system that uses keyword filtering and

logistic regression to detect crisis messages in existing clinical telehealth chat workflows. By flagging high-risk messages in real time, the system enabled humans to triage crisis messages.

AI systems can also assist in detecting disclosures of violence on social media platforms, where survivors may share their experiences anonymously. Al-Garadi et al. (2022) designed an automated system using NLP and machine learning to detect IPV-related content on Twitter, now known as X. Their model, which is based on Robustly Optimized BERT (RoBERTa), achieved an F1 score of 0.76 in classifying personal IPV reports compared to other models, highlighting how social media can supplement traditional detection pathways. Although this tool is currently in research stages, Al-Garadi's team calls for the practical applications of this effective, low-bias, and reliable model to classify self-reported IPV on social media, which could lay the foundation for designing evidence-based interventions, supporting IPV victims, and potentially addressing IPV in near real time through social media.

Although primarily designed for individual-level detection, such tools may also contribute to broader surveillance efforts when applied at scale over time—an area discussed further in the next section. Thanks to these tools, we can see undisclosed violence, but the further question is: are these risks expanding? Are they concentrated in certain populations or time periods?

### 2.1.2 Risk Surveillance: Tracking Population-Level Trends and Patterns

While risk detection focuses on identifying individual instances of violence, risk surveillance refers to the systematic, ongoing use of AI tools to observe patterns, fluctuations, and disparities in S/IPV incidents at the population level. These tools and evidence help public health professionals and policymakers gain insight into how violence evolves across time and thus inform public health or policy responses.

One example of AI-enabled surveillance is the GBVMeter tool developed by Manzoor et al. (2022), which analyzed 12,693 English-language news articles on gender-based violence (GBV) across Pakistan, India, and the UK. The researchers applied Latent Dirichlet Allocation (LDA) for topic modeling and introduced a word embedding technique, GBV2Vec, built on Doc2Vec, to classify and assess the sensitivity of GBV-related content. Their analysis found that rape was the most frequently reported form of GBV, followed by domestic violence, violence against women, and sexual violence. In a case study during the COVID-19 pandemic, results indicated a global increase in GBV, with women disproportionately affected. By identifying reporting patterns, topic salience, and narrative shifts during crisis periods, the GBVMeter tool provides actionable insights for policymakers and human rights advocates, enabling timely interventions and advancing Sustainable Development Goals (SDGs) such as gender equality and inclusive societies.

Another example comes from Hodgkinson et al. (2023), who analyzed 43,488 domestic abuse crime records from a UK police force during the COVID-19 pandemic. Using NLP to extract insights from free-text fields and combining

change-point analysis with anomaly detection, the study tested hypotheses about reporting behavior, cohabitation effects, and timing of abuse trends. Contrary to expectations, the data showed no increase in domestic abuse during the first national lockdown, but a significant and sustained rise after restrictions were lifted. Notably, the increase did not appear to reflect changes in reporting rates or cohabitation patterns. These findings demonstrate how AI-enhanced surveillance can reveal complex, time-sensitive patterns in administrative data, thereby informing more adaptive, evidence-based policy responses.

In another example, Chen et al. (2023) addressed the challenge of underreported IPV prevalence by applying machine learning to the Third Wave Survey on the Social Status of Women in China. Using a random forest algorithm to impute missing data, they found significantly higher rates of physical (7.10%), verbal (13.74%), and cold violence (21.35%) than in traditional survey responses (4.05%, 11.21%, and 17.95%, respectively). Here, cold violence is an emotional form of domestic violence characterized by a complete withdrawal of all verbal and physical communication by one member of an intimate relationship toward the other (McLaren, 2016). This highlights how AI can enhance surveillance accuracy by estimating the true prevalence of sensitive social issues and correcting for non-response bias.

These examples highlight how AI can help answer key public health questions: Where does violence occur? Is it increasing or decreasing? Which groups are most at risk? Importantly, surveillance systems are not intended to identify individual victims or intervene directly, but rather to inform service planning, funding allocation, and prevention programming. When applied ethically and transparently, AI-supported surveillance enables a shift from reactive responses to a proactive strategy, empowering institutions to recognize macro-level signals and respond with data-informed policies. The following section builds on this by exploring how AI can move even further upstream: predicting who might be at greatest risk and intervening before violence occurs.

### 2.1.3 Risk Prediction: Survivor and Perpetrator

AI-based risk prediction tools aim to estimate the likelihood of future S/IPV victimization or perpetration by identifying patterns in population-level historical data. Unlike detection and surveillance tools that focus on identifying existing incidents or tracking trends, predictive models are typically used in research contexts to explore population-level risk factors and inform prevention-oriented strategies. These tools are not intended for individual-level decision-making in clinical or legal settings, and their use raises important ethical considerations, including privacy, consent, and the agency of survivors and at-risk individuals.

From a survivor-centered perspective, researchers have explored the use of predictive models to improve early identification of individuals who may be at heightened risk of severe harm. For example, Garcia-Vergara et al. (2023) trained a machine learning classifier using legal documents to distinguish between lethal and non-lethal IPV cases, achieving an accuracy of 83.16%. The goal of this model was

not to label individuals, but to support risk stratification in research that could inform resource allocation or the design of lethality screening tools. However, any application of predictive models to identify potential victims must be approached with caution. Survivors may choose not to disclose abuse for valid and personal reasons, and systems that infer risk without their consent risk undermining their agency and privacy. Therefore, current models are developed with de-identified, aggregate data, and their outputs are used for population-level insights, not individual monitoring or intervention.

Recent studies have also explored predicting violent behavior from the perpetrator's perspective, building on the intergenerational transmission of violence theory (Widom & Wilson, 2015) derived from Bandura's (1977) social learning theory. This framework posits that exposure to violence during childhood, either as a victim or witness, increases the likelihood of using violence as a conflict resolution tactic in adulthood. Childhood maltreatment or abuse is identified as the most common predictor of youth intimate partner violence (IPV) perpetration. Petering et al. (2018) examined this phenomenon among homeless youth, aiming to identify key variables that determine the likelihood of IPV perpetration and developed an AI-based triage tool to predict such behavior. The study used supervised machine learning algorithms to analyze data from 353 homeless youth, focusing on physical perpetration ranging from minor (e.g., slapping) to severe (e.g., kicking). Key findings indicate that PTSD, depression, prior violence engagement, hard drug use, and loneliness are among the most influential predictors of IPV perpetration. Support Vector Machines (SVMs) with a Radial Basis Function (RBF) kernel demonstrated superior performance compared to other algorithms (e.g., Random Forests and Neural Networks), achieving the highest F1-score (0.53) and ROC AUC (0.69). However, it is essential to clarify that such models are currently limited to research contexts, often focused on high-risk or already-identified populations (such as homeless youth), rather than being broadly applied to all youth or used in routine screening. Their main contribution is in identifying population-level risk patterns and supporting the development of early prevention strategies, rather than labeling or profiling individuals for intervention. In practice, these tools are not intended for practitioners to use directly to screen or label individual perpetrators in real-world settings. Instead, they help inform policy, shape targeted interventions, and expand our understanding of risk pathways. The use of such models raises critical ethical questions, particularly around privacy, potential bias, and the dangers of misclassification or stigmatization if predictive results are misapplied outside research environments. Responsible implementation demands clear boundaries, robust data governance, and ongoing ethical oversight to prevent unintended harm.

While perpetrator-focused AI modeling shows promise for research and program design, practical deployment for case-level decision-making remains premature. Continued evaluation, transparency, and co-design with stakeholders are necessary to ensure these tools empower, rather than endanger, those most affected by IPV.



## 2.2 *AI Tools for Assessment and Intervention*

According to World Health Organization guidelines (WHO, 2013), immediate support (first-line support) is strongly advised for women disclosing sexual or intimate partner violence (S/IPV). A systematic review (Novitzky et al., 2023) classifies S/IPV interventions into three levels: primary, secondary, and tertiary. Primary prevention targets reducing health threats before they occur, typically via school, community, and digital platforms aimed at adolescents and families to tackle the root causes of IPV. Secondary prevention focuses on early intervention post-exposure, encompassing IPV screening, shelter relocation, counseling, legal support, and perpetrator interventions such as diversion programs and anger management. Tertiary prevention addresses the long-term impacts of past IPV experiences, emphasizing psychological support, recidivism reduction, and community reintegration.

Accurate recognition of survivors' specific needs is fundamental for effective intervention. Traditional screening tools rely on fixed checklists or manual triage, which can miss or delay the nuanced, rapidly changing requests survivors make online or during clinical encounters. Recent large-language-model (LLM) studies show that AI can bridge this gap by classifying survivors' self-described information needs in real time. Hui et al. (2024) fine-tuned a large language model to successfully predict 66.6% of the informational needs of domestic violence victims from online posts on Reddit. Guan et al. (2025) recently improved this prediction model to 2216 posts using GPT-3.5-generated synthetic examples and achieved an F1-score of 0.705 across eight need categories (e.g., shelters, legal aid, childbearing, safety planning, DV knowledge, etc.). These findings have the potential to be translated into practice. Guan et al. (2025) outlined how such LLMs could be embedded in electronic health-record portals to trigger real-time, need-matched recommendations, enabling community nurses, social workers, and counselors to deliver personalized resources without delay. These findings have the potential to be translated directly into clinical practice: when embedded in electronic health-record portals or NGO chatbots, LLM-based triage can automatically surface tailored resources, field routine queries, and flag ambiguous or high-risk messages, enabling nurses, social workers, and counselors to focus their expertise where it matters most. The authors also emphasize that development and deployment must be co-designed with survivors and service agencies to ensure all interventions remain trauma-informed, culturally appropriate, and ethically sound.

AI also contributes significantly to assessing the severity of violence, ensuring fairness and consistency in evaluations. In a study by Sikstrom et al. (2023), researchers compared human and machine assessment of severity in physical and psychological IPV events. Participants who experienced violence rated its severity, while another group assessed the same events. Results showed that human evaluations were affected by bias, particularly underestimating the severity of psychological violence. In contrast, a computational model trained on word embeddings and supervised learning not only achieved higher predictive accuracy but also mitigated calibration bias between self-report and observer ratings. These findings suggest



that AI tools could support legal decision-making, prioritize services, and ensure fair media representation—particularly in domains where violence is often subjectively judged.

In addition to identifying and assessing survivors' needs, technology can enhance the scalability and accessibility of interventions. Al-Alosi (2020) outlines five key areas where technology can play a positive role: delivering resources and services, combating loneliness and isolation, supporting safety planning, documenting abuse, and empowering victims. AI in this context emerges as a promising technique to enhance traditional digital service delivery tools.

One application is assisting healthcare providers through AI-enabled triage systems. For instance, Swaminathan et al. (2023) developed an NLP-based model that can reliably identify crisis-related messages (including domestic violence and self-harm), decreasing the median triage time from 9 h to 8–13 min. This integration demonstrates AI's practical utility in streamlining clinical workflows and promptly addressing survivors' urgent needs.

Another application is direct client-facing chatbots. The current S/IPV-targeted AI chatbots include Aimee Says, Keepit, and ChattyCuz. Aimee Says, a web-based chatbot, has supported over 70,000 domestic violence survivors by offering confidential guidance, legal information, and safety planning referrals. It securely documents abuse incidents, enabling survivors to navigate legal processes safely and privately. Despite its effectiveness, the platform acknowledges limitations in handling deeply emotional or sensitive issues, reinforcing the ongoing need for human intervention (Gambino, 2024). Similarly, Keepit, developed by Nigerian activist Bukky Shonibare, leverages NLP, machine learning, and cloud computing to offer a confidential, empathetic space where survivors of sexual violence can safely document their experiences. It aims to empower survivors by preserving their testimonies for potential legal actions, providing emotional support, and facilitating evidence collection, legal preparation, and referrals. Recognizing that survivors may prefer interacting without being judged, Keepit ensures survivors' comfort and autonomy during sensitive disclosures (Segal, 2025).

While there is limited research on the effectiveness and safety of models like Aimee Says and Keepit, an empirical intervention study (De Filippo et al., 2023) highlights the potential of a behaviorally informed gamified chatbot called ChattyCuz (a WhatsApp chatbot intervention) as a low-cost intervention to improve gender-equitable attitudes and reduce intimate partner violence (IPV) exposure among young women in South Africa. The research examined the efficacy of behaviorally informed digital chatbots, *Attention Treatment*, *Gamified Treatment*, and *Narrative Treatment*, in enhancing gender-equitable attitudes and reducing IPV exposure among South African young women. All chatbot interventions improved gender attitudes, with the gamified version showing the most significant effect. This gamified chatbot also modestly yet significantly reduced the experience of past-month IPV exposure, especially technological and psychological IPV, with an 8% decrease as reported by women. Moreover, it enhanced users' ability to identify unhealthy relationship behaviors. Despite no impact on depressive symptoms, the chatbots' safety was ensured with a counselor link. Although promising, the

researchers emphasize the need to expand future studies beyond heteronormative frameworks and single-platform contexts to strengthen generalizability and applicability.

These AI-based tools illustrate transformative potential in IPV interventions, shifting from passive support to active, personalized, and scalable strategies. Ethical implementation, rigorous evaluation, and survivor-centered design remain critical as technology increasingly shapes IPV intervention paradigms.

### ***2.3 Survivor Empowerment***

As described in previous sections, AI tools have transformed the landscape of S/IPV response by enhancing both provider-driven interventions and direct survivor access to support. Yet, the most far-reaching potential of AI in this context lies in its capacity to empower survivors. In the context of S/IPV, a “survivor” is an individual who has lived through and overcome experiences of sexual and intimate partner violence. Survivor empowerment is a cornerstone of effective responses to IPV. Empowerment, as defined by Dutton’s empowerment theory (Dutton, 1992), recognizes the survivor’s inherent strength and aims to equip them with the choice, knowledge, skills, resources, and sense of control necessary to ensure their safety, promote their healing, and rebuild their lives.

First, AI significantly enhances survivors’ autonomy by making critical information and resources more accessible, timely, and personalized. Research by Hui et al. (2024) and Guan et al. (2025) demonstrated the capability of fine-tuned large language models to classify survivors’ expressed informational needs from online disclosures accurately. Such AI applications would provide survivors with immediate, confidential access to tailored information, empowering them to independently navigate their help-seeking journey without stigma, judgment, or logistical barriers inherent to traditional service environments.

Second, AI interventions contribute to survivors’ psychological resilience, self-efficacy, and proactive safety planning. Technology-based interventions, potentially incorporating AI elements, can provide discretionary and reliable support, psychoeducation, and information that empowers survivors to make informed decisions about their safety and well-being (Emezue et al., 2022). These interventions can be tailored to specific populations, such as immigrant women, to address their unique needs and increase their sense of agency (Sabri et al., 2021). Features can include information on relationship red flags, danger assessment, and personalized safety plans with links to local and national resources (Emezue et al., 2022; Sabri et al., 2021). As Aimee Says, Keepit, and ChattyCuz, these AI chatbots are designed for direct survivor interaction, providing confidential, judgment-free spaces for users to share experiences, seek guidance, and receive support at any hour. These digital assistants enable survivors to document incidents securely, preserve testimony for potential future legal action, and engage in conversational scenarios that build confidence in recognizing and responding to unhealthy relationships. By lowering the

barriers to disclosure, offering step-by-step safety planning, and connecting users to relevant resources without the need for face-to-face interaction, such chatbots expand the autonomy and reach of support for those who may be hesitant, isolated, or marginalized. Nevertheless, these interventions are most effective when complemented by human oversight and are developed with survivor feedback to ensure ethical, culturally sensitive, and trauma-informed care.

Third, AI can reduce structural barriers and foster equitable access to support for marginalized or underserved populations. For example, Tabaie et al. (2022) demonstrated that NLP techniques analyzing electronic medical record notes could identify IPV cases with much greater precision than traditional ICD code-based methods. In practice, these AI systems are most often integrated into healthcare providers' electronic record workflows, where they flag potential IPV indicators for clinicians, social workers, or case managers. This allows healthcare professionals to proactively offer survivors referrals to local resources, legal aid, or counseling that they might not otherwise access—directly aligning with the World Health Organization's recommendations for connecting survivors to community services (WHO, 2013).

Drawing on the AI intervention tools discussed in Sect. 2.2, we can envision that the pathway to improved community resource access can be both healthcare-provider- and survivor-driven. In healthcare provider-facing models, AI serves as a decision-support tool: once a possible IPV case is identified, clinicians are prompted to initiate sensitive conversations and refer patients to appropriate services, with real-time, locally tailored information supplied by the AI system. Alternatively, as patient-facing tools become more common, survivors themselves may use AI-enabled portals to learn about confidentiality or to request services covered by their healthcare plans or available in their communities. In both models, the value of AI lies in its ability to deliver up-to-date, individualized, and human-reviewed recommendations, ensuring that survivors are matched with accurate, contextually relevant resources. As large language models evolve and “human-in-the-loop” designs mature, these systems will increasingly bridge the gap between detection and actionable support, making equity and personalized access to resources a practical reality for more survivors.

### 3 Ethical Considerations in AI Applications

The intersection of AI with S/IPV presents complex ethical challenges that demand careful consideration. Current literature reveals significant concerns regarding privacy violations through AI-enabled deepfakes, the potential for technology to exacerbate trauma, and risks of misidentification that can perpetuate harm. Research indicates that malicious AI applications can create realistic, indistinguishable fabricated content, while AI-based re-identification attacks render traditional anonymization measures increasingly ineffective. Moreover, emerging evidence suggests that trauma-informed approaches in S/IPV-related AI design are essential to prevent technology from becoming another vector for abuse. This section discusses the

ethical considerations around privacy violations, the risk of retraumatization, bias, and a lack of transparency. It underscores the importance of ethically deploying AI to empower survivors and enhance support services through trauma- and violence-informed approaches, highlighting both the evolving threats and potential protective frameworks.

### ***3.1 Privacy Violations from General AI Technologies***

AI technologies have enabled unprecedented capabilities for creating fabricated content that violates privacy and dignity. Deepfakes, which are “images or recordings that have been convincingly altered and manipulated to misrepresent someone as doing or saying something that they did not actually do or say,” have become particularly problematic in the context of sexual abuse (Okolie, 2023). AI technologies are extensively misused in the pornography industry, where women’s faces are digitally mapped onto other bodies to create non-consensual sexual content. This form of technology-facilitated abuse constitutes a severe privacy violation with lasting psychological impacts on victims. The hyper-realism of AI-generated materials makes them nearly indistinguishable from authentic content, amplifying their harm. Moreover, the permanence and replicability of AI-generated content allow this abuse to persist indefinitely, perpetuating trauma and intensifying long-term psychological effects such as PTSD, depression, and reduced personal security (Abdulai, 2025). Another research (Hawkes et al., 2024) highlights the vulnerability of image-based sexual abuse removal tools to generative AI attacks, which can approximately reconstruct original image material. This result emphasizes the need for stronger data protection measures, such as cryptographic protocols like private set intersection (PSI). By enabling secure hash matching without exposing sensitive data, PSI offers a privacy-focused solution for removing harmful content and safeguarding vulnerable users. This approach not only counters adversarial AI attacks but also enhances privacy protection in combating technology-facilitated abuse.

Beyond the creation of fabricated content, AI systems present substantial privacy risks through person re-identification (Platzer, 2022). Traditional data masking and anonymization approaches have proven increasingly vulnerable to sophisticated AI algorithms that can re-identify individuals based on behavioral patterns. This concept has attracted increasing attention from the research community due to its vital role in security applications such as tracking robberies and preventing terrorist attacks (Yadav & Vishwakarma, 2023). However, the implications for survivors of S/IPV are particularly concerning, as maintaining privacy and confidentiality can be essential for safety. If S/IPV abusers can leverage AI re-identification techniques to locate or gather information about survivors, this technological vulnerability becomes a physical safety threat.

In addition to the intentional use of malicious AI models, general AI technologies may increase the risk for technology-mediated trauma in two other ways. First, the training of AI algorithms often relies on underrepresented, biased, and

inaccurate datasets, which can perpetuate harm. Second, general AI-powered chatbots may provide inappropriate health advice, further endangering vulnerable populations (Abdulai, 2025). Thus, general AI technologies may expand avenues for sexual exploitation and abuse, but also are not suitable for supporting survivors of S/IPV without careful ethical consideration and safeguards.

### ***3.2 Ethical Challenges in AI Tools Designed for S/IPV Response***

Following the discussion of general-purpose AI misuse in S/IPV contexts (Sect. 3.1), this section shifts focus to the distinct ethical challenges associated with AI tools intentionally developed for S/IPV prevention and survivor support. While AI tools designed for S/IPV response (AI-S/IPV) hold promise for preventing S/IPV and survivor empowerment, as stated in Sect. 2, the absence of clear ethical governing principles and implementation guidelines poses a significant hurdle. Professionals lack comprehensive guidance on how to deploy these emerging technologies responsibly in such a sensitive domain.

First, the risk of retraumatization is a critical concern. Generative AI in digital health interventions could inadvertently trigger, perpetuate, or exacerbate prior traumatic experiences among vulnerable populations and could widen health inequities (Abdulai, 2025). Tabaie et al. (2022) acknowledge the potential for survivor retraumatization if the identification and subsequent interventions related to IPV are not handled sensitively.

Second, bias and fairness issues arise from the datasets used to train AI/ML models. These datasets often exhibit a bias toward gender-based violence against women, potentially neglecting other forms of IPV (Novitzky et al., 2023). This bias can skew the effectiveness of interventions and marginalize victims whose experiences fall outside the narrow scope of the training data, thereby undermining the equity and inclusivity of these technologies.

Third, the lack of transparency and explainability presents a significant trust challenge. Crooks et al. (2024) highlight the challenge of building trust in AI output, particularly when the AI relies on opaque techniques. If users, including survivors and healthcare providers, do not understand how an AI arrived at a recommendation or action, they may not trust or utilize it effectively. This lack of transparency can be an ethical concern, especially in sensitive contexts like S/IPV, where understanding the rationale behind AI interventions is essential.

These interconnected ethical challenges call for a comprehensive and responsible approach to the development and deployment of AI technologies in the domain of S/IPV. Addressing these challenges is essential to ensure that AI interventions are not only effective but also ethical and empowering for survivors.

### ***3.3 Trauma- and Violence-Informed Approaches in AI-S/ IPV Development***

As recommended by the WHO, a trauma-informed approach is essential for S/IPV services. Traditional trauma-informed care (TIC) emphasizes that trauma affects all aspects of a person's life—including families and communities—and that providers must recognize trauma's signs, respond sensitively at both the individual and organizational level, and actively resist retraumatization (Palmieri & Valentine, 2021).

Trauma- and violence-informed care (TVIC) expands upon these foundations by explicitly addressing the role of structural violence and systemic inequities in survivors' lives (Wathen & Mantler, 2022). TVIC is guided by four core principles: (1) understanding and awareness of trauma and violence, especially structural violence, and their impacts; (2) prioritizing the physical, emotional, and cultural safety of all involved; (3) fostering person-centered connection, collaboration, and choice; and (4) building on individuals' strengths and supporting their skill and capacity development (Wathen & Varcoe, 2023). This paradigm shift is crucial, as it places responsibility on organizations and systems, not just individuals, to adapt services, policies, and environments for survivors, rather than expecting survivors to adjust to rigid or potentially harmful systems.

When considering integrating AI into S/IPV interventions, embedding TVIC principles is both a challenge and an opportunity. Conventional support systems often require survivors to formally disclose IPV and retell their experiences multiple times to access care, a process that can perpetuate harm and retraumatization (Wathen & Mantler, 2022). AI-enabled tools, especially the chatbots introduced in the previous sections, can help address this by allowing survivors to securely share their story once with informed consent and then facilitate the transfer of necessary information across authorized providers as needed. This approach aligns closely with TVIC's emphasis on minimizing retraumatization, safeguarding privacy, and maximizing survivor choice and autonomy.

To fully realize these benefits, AI tools for S/IPV response must be designed, implemented, and evaluated through a TVIC lens. This means prioritizing survivor safety, empowerment, and cultural sensitivity throughout the AI development life-cycle and ensuring that the technology is adaptable to diverse needs and contexts. TVIC also demands that organizations proactively address systemic and institutional barriers, including policies or digital practices that might inadvertently perpetuate inequity or harm.

Recent innovations demonstrate the need for robust ethical governance. For instance, a team at the University of Queensland has initiated the co-design of an AI trauma-informed audit framework to assess the extent to which AI-driven decisions may cause new trauma or retraumatization. Notably, such frameworks should inform the design of digital interventions from inception, rather than only retrospectively assessing harm after deployment. While the details of this particular framework are yet to be published, its development signals a promising direction for future AI governance.

Another forward-thinking principle is “Safety-by-Design” (Thorn & All Tech Is Human, 2024). They indicate that proactive approaches centered on safety-by-design principles are essential for mitigating potential harm. While this particular reference focuses on child sexual abuse, the principle could extend to adult survivors of S/IPV. The safety-by-design frameworks should be “tactical, actionable, and multidisciplinary,” enabling technical, policy, product, and trust and safety teams to implement protective measures with minimal friction. These approaches represent a shift from reactive responses to harm toward proactive prevention of technology-facilitated abuse. This shift is fundamental given that the “prevalence of AI-generated child sexual abuse material is small, but growing,” suggesting a critical window for intervention before these problems become more widespread.

As AI becomes increasingly integrated into S/IPV support, grounding technology-mediated interventions in TVIC and related frameworks is essential for advancing equity, effectiveness, and survivor well-being. Embedding these principles, supported by strong organizational commitment and ongoing ethical reflection, will help ensure that digital innovation empowers rather than inadvertently harms those it is designed to serve.

The literature on ethical considerations around AI applications related to S/IPV reveals several insights. First, AI technologies present significant privacy threats through deepfakes, re-identification attacks, and other forms of technology-mediated trauma. These threats can perpetuate trauma and create ongoing safety risks for survivors. Second, AI systems face challenges of bias and a lack of transparency. Biased datasets can reduce intervention effectiveness and marginalize some victims. Also, AI’s unclear decision-making process makes it hard to trust, especially in S/IPV cases where understanding AI’s logic is key. This shows the need for ethical rules that focus on fairness and accountability in AI use. Third, implementing trauma- and violence-informed approaches to AI design and deployment is essential but currently underdeveloped. Safety-by-design principles offer a new framework for proactively preventing harm, but they must be tailored to address the vulnerabilities of individuals experiencing S/IPV.

## 4 Discussion and Future Directions

This chapter has explored the transformative opportunities that AI brings to the field of sexual and intimate partner violence (S/IPV) prevention and survivor support. Rather than serving as a replacement for established practices, AI offers new tools to augment and diversify approaches for detection, intervention, and survivor empowerment. However, as the preceding sections make clear, integrating AI into S/IPV services cannot be straightforward or unproblematic.



## ***4.1 Critical Reflections on the Current Evidence Base***

The literature reviewed in this chapter demonstrates promising use cases for AI, including NLP for risk detection, large language models for triaging information needs, and chatbots for preliminary support. Yet, much of the available evidence remains limited in scope. Most AI-enabled interventions have only been evaluated as research prototypes or in narrowly defined pilot settings, often with modest sample sizes and limited demographic diversity. For example, while studies have shown that machine learning models can predict risk factors for perpetration among specific high-risk groups, these findings are not easily generalizable to broader populations or to routine practice. Furthermore, there is a lack of longitudinal and real-world impact evaluations.

Another key limitation lies in the conceptual framing of risk and intervention. Some AI models, in their attempt to “predict” or classify risk, may inadvertently shift the focus onto individual behaviors or characteristics, potentially reinforcing the misperception that survivors are responsible for their victimization or that perpetrators can be reduced to a set of risk factors. Such framings risk obscuring the intersectional drivers of S/IPV. It is essential for future research to adopt frameworks that respect survivors and realities, and to avoid reinforcing stigma or blame.

## ***4.2 Ongoing Challenges in Implementation***

Despite the rapid evolution of AI technologies, integrating AI into real-world S/IPV services faces formidable hurdles. Most current applications, such as AI chatbots, have not yet achieved seamless linkage with community resources, legal advocacy, or healthcare systems. Initiatives like Aimee Says demonstrate the potential for digital support, but more work is needed to ensure that these tools are embedded within a holistic, trauma- and violence-informed care ecosystem. Projects such as Keepit, while innovative, have yet to move beyond development and into sustainable practice.

Additionally, unresolved concerns remain regarding data privacy, informed consent, bias mitigation, and the risk of retraumatization. The “last mile” problem, ensuring that AI-generated insights translate into tangible, survivor-centered action within complex social systems, remains acute. Service providers must be vigilant to ensure that the introduction of AI does not inadvertently widen disparities or introduce new barriers, particularly for marginalized populations.

### ***4.3 Future Directions for Research, Practice, and Policy***

For researchers and developers:

- Foster interdisciplinary and multi-institutional collaboration to design and validate AI tools grounded in real-world needs and trauma- and violence-informed care (TVIC) principles.
- Conduct large-scale, longitudinal, and cross-cultural studies to ensure AI interventions are inclusive, robust, and responsive to the diverse experiences of survivors.
- Embrace participatory, survivor-centered co-design at every stage of development, actively addressing bias, privacy, consent, and the risk of retraumatization.

For practitioners and service providers:

- Build capacity to leverage AI tools as complements to professional judgment, emphasizing human–AI collaboration in screening, triage, and resource navigation.
- Uphold survivor choice, informed consent, and respect throughout AI-assisted workflows, and contribute real-time feedback to improve tool safety and relevance.
- Build institutional standards for ethical review, data security, and trauma- and violence-informed practice, ensuring technology always serves the safety and empowerment of those affected by S/IPV.

For policy-makers and regulators:

- Develop and enforce ethical, privacy, and accountability standards for AI use in S/IPV settings, prioritizing transparency and “safety-by-design” principles.
- Support and scale up multi-sector pilot projects that model collaborative, survivor-centered AI implementation.
- Promote equitable access and funding for inclusive AI solutions, ensuring that innovations do not deepen existing disparities or exclude marginalized communities.
- Establish ongoing mechanisms for feedback, monitoring, and policy adaptation as technologies and service needs evolve.

In summary, AI presents a transformative opportunity for S/IPV researchers, service providers, policymakers, and survivors alike. Its responsible integration demands a commitment to rigorous evidence, ethical reflexivity, survivor-centered design, and collaborative, multidisciplinary practice. As technology continues to advance, the field must remain vigilant to ensure that innovation serves not just the promise of efficiency, but the reality of safety, dignity, and empowerment for all survivors.

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